

Library Book Borrowing System

End-of-Semester Dynamic Web Application Project - Option E

System Description

The Library Book Borrowing System is a PHP and MySQL web application for managing student book loans. Students can register, log in, search the catalogue by title or author, borrow available books, return books, and view their borrowing history. Admin users can manage book records and monitor borrowing activity.

Functional Requirements Covered

Requirement	Implementation
Registration, login, logout, and sessions	`User.php`, `register.php`, `login.php`, and `logout.php` manage authentication and protected pages.
Forms, validation, and persistent data	Student registration, book borrowing, returns, and admin book forms store data in MySQL with feedback.
Dynamic display, search, and filtering	Catalogue, borrowing history, admin dashboard, and borrowing activity pages retrieve records from MySQL.
Edit, delete, and role-based access	Admins can add, edit, and delete books. Admin pages require the `admin` role.

Database Structure

Table	Main Columns	Purpose
users	user_id, username, email, password, full_name, role, enrollment_id, phone, address, is_active	Stores student and admin accounts.
books	book_id, title, author, isbn, publisher, publication_year, category, total_copies, available_copies, description	Stores book catalogue records and availability.
borrowing_records	record_id, user_id, book_id, borrow_date, due_date, return_date, status, fine_amount	Stores borrowing and return transactions.

Key Screens

Library Book Borrowing System

LoginRegister

Welcome to Library Book Borrowing System

Manage your book borrowing experience

LoginRegister

1. Home page

Library System

HomeRegister

Student Login

Username

Password

Login

Don't have an account? Register here

2. Login page

Library System

HomeCatalogueMy BorrowesHistoryLogout

Book Catalogue

Search by title or author

Search

1984

Author: George Orwell

ISBN: 978-0451524935

Category: Dystopian

Available: 2 / 2

A dystopian social science fiction novel...

Pride and Prejudice

Author: Jane Austen

ISBN: 978-0141439518

Category: Romance

Available: 4 / 4

A romantic novel of manners...

The Catcher in the Rye

Author: J.D. Salinger

ISBN: 978-0316769174

Category: Fiction

Available: 2 / 2

A story of teenage rebellion and alienation...

3. Student book catalogue

Library System

HomeCatalogueMy BorrowesHistoryLogout

My Borrowing History

Book Title	Author	ISBN	Borrowed Date	Due Date	Returned Date	Status	Fine
The Great Gatsby	F. Scott Fitzgerald	978-0743273565	May 05, 2026	May 19, 2026	May 05, 2026	Returned	-

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4. Student borrowing history

Library System - Admin Panel

HomeDashboardManage BooksBorrowing ActivityLogout

Manage Books

+ Add New Book

Books List

Title	Author	ISBN	Category	Total	Available	Actions
1984	George Orwell	978-0451524935	Dystopian	2	2	Edit Delete
Pride and Prejudice	Jane Austen	978-0141439518	Romance	4	4	Edit Delete
The Catcher in the Rye	J.D. Salinger	978-0316769174	Fiction	2	2	Edit Delete

5. Admin book management

Library System - Admin Panel

HomeDashboardManage BooksBorrowing ActivityLogout

Borrowing Activity

[All](#) [Active Borrowers](#) [Overdue Books](#)

Student	Email	Book Title	Author	Borrow Date	Due Date	Returns Date	Status	Fine	Action
Demo Student	student@library.com	The Great Gatsby	F. Scott Fitzgerald	May 05, 2026	May 19, 2026	May 05, 2026	Returned	-	-

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6. Admin borrowing activity

Challenges and Lessons Learned

A key challenge was coordinating borrowing records with book availability so that returned books become available again and active loans are protected from accidental deletion. Another challenge was keeping student and admin permissions separate while still allowing both roles to use the system smoothly.

I learned how to connect PHP pages to a MySQL database using prepared statements, manage user sessions, validate form input, and organize a dynamic web application into reusable classes for users, books, borrowing records, and database access.

Conclusion

The finished system meets the required capabilities for a dynamic database-backed web application. It includes authentication, protected pages, role-based access, data entry, validation, dynamic display, search, editing, deletion, and persistent MySQL storage.